DSA Practical No: 09

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Prims Algorithm

INPUT:

```
#include <limits.h>
#include <stdbool.h>
#include <stdio.h>

#define V 5

int minKey(int key[], bool mstSet[])
{
  int min = INT_MAX;
  int min_index = -1;

  for (int v = 0; v < V; v++)
  {
    if (mstSet[v] == false && key[v] < min)
    {
      min = key[v];
      min_index = v;
    }
  }

  return min_index;
}
```

```
void printMST(int parent[], int graph[V][V])
{
   int total = 0;

   printf("\nEdge\tWeight\n");

   for (int i = 1; i < V; i++)
   {
      printf("%d - %d\t%d\n",
            parent[i], i, graph[parent[i]][i]);

      total += graph[parent[i]][i];
   }

   printf("Total weight of MST = %d\n", total);
}

void primMST(int graph[V][V])
{
   int parent[V];
   int key[V];
   bool mstSet[V];

   for (int i = 0; i < V; i++)
   {
      key[i] = INT_MAX;
      mstSet[i] = false;
```

```
    parent[i] = -1;
}

key[0] = 0;

for (int count = 0; count < V; count++)
{
    int u = minKey(key, mstSet);

    if (u == -1)
    {
        printf("Graph is disconnected. MST cannot be formed.\n");
        return;
    }

    mstSet[u] = true;

    for (int v = 0; v < V; v++)
    {
        if (graph[u][v] != 0 &&
            mstSet[v] == false &&
            graph[u][v] < key[v])
        {
            parent[v] = u;
            key[v] = graph[u][v];
        }
    }
}
```

```
    printMST(parent, graph);
}

int main()
{
    int graph[V][V] = {
        {0, 2, 0, 6, 0},
        {2, 0, 3, 8, 5},
        {0, 3, 0, 0, 7},
        {6, 8, 0, 0, 9},
        {0, 5, 7, 9, 0}
    };

    primMST(graph);

    return 0;
}
```

OUTPUT:

```
PS D:\ENGINEERING\SEM I\DATA STRUCTURE\Experiment-09-Prims-Algorithum> .\prim.exe

Edge    Weight
0 - 1   2
1 - 2   3
0 - 3   6
1 - 4   5
Total weight of MST = 16
PS D:\ENGINEERING\SEM I\DATA STRUCTURE\Experiment-09-Prims-Algorithum>
```